

Take-Home Assignment

Junior AI Automation Engineer · Internship / Entry-Level · Bangalore

Estimated time	3–4 hours (designed to finish in ~2)
Submission	Public GitHub repo + README + sample_output/ folder
Deadline	5 working days from receipt

The Scenario

You're joining the ops team at a logistics company. Every day, ~20 vendor emails arrive with attached invoices (PDFs or scanned images). The team manually reads each one, classifies it, extracts key fields, and routes it — high-value invoices go to a manager Slack channel, others are logged to a spreadsheet and auto-acknowledged.

Your task: Build an end-to-end AI agent pipeline that handles this entirely — from raw email input to routed output — with no human in the loop.

What to Build — 4 Stages

1

Ingestion & Document Parsing

Ingestion

Accept a webhook payload (or a local folder of files) simulating incoming vendor emails. Each 'email' has a sender, subject, and one attached file — a PDF invoice or image. Parse the attachment and extract its raw text. You may use any OCR library, LLM vision call, or PDF parser of your choice.

2

Classification & Field Extraction

Extraction

Using an LLM API, classify each document as one of: standard_invoice, credit_note, or unknown. Then extract the following fields: vendor name, invoice number, date, line items, and total amount. Output must be a validated, structured JSON object per document — not free-form text.

3

Conditional Routing

Routing

Apply routing rules: if total > Rs.50,000 — post a formatted Slack message (real or mock webhook). If total <= Rs.50,000 — append a row to Google Sheets or a local CSV as fallback. If classified as unknown — write to a separate human_review log. No document should be silently dropped.

4

Acknowledgement & Error Handling

Error Handling

For each processed document, log a structured outcome: success, partial, or failed, with a reason string. Send a mock acknowledgement email back to the vendor (console log or real send via SendGrid free tier). Demonstrate graceful degradation: what happens when the LLM returns malformed JSON? When an attachment is unreadable?

Sample Data Provided

The following test files will be emailed to you alongside this brief. A mock webhook payload JSON is included so you don't need a live mail server.

- 5 × Standard invoices (varying formats, fonts, layouts)
- 2 × Credit notes
- 2 × Scanned image invoices (JPG, varying quality)
- 1 × Intentionally messy / partially unreadable file

Note: At least 2 invoices exceed Rs.50,000 to test the Slack routing branch. At least 1 invoice has an ambiguous total (subtotal vs GST-inclusive) — handle it deliberately and explain your choice in the README.

Evaluation Rubric

Dimension	Weight	What we look for
Core pipeline works	30%	End-to-end flow runs on sample data without manual intervention
Extraction accuracy	20%	JSON output matches ground truth for 8+ of 10 docs; output is validated
Routing correctness	15%	Slack vs Sheet follows rules; unknown docs flagged, not silently dropped
Error & edge case handling	15%	Malformed LLM output is caught and retried or flagged; bad files produce structured error logs
Code quality	10%	Readable, modular — not spaghetti. Secrets in env vars, not hardcoded
Documentation	10%	README covers setup, run instructions, design decisions, known limits — under 1 page

Bonus — No Extra Marks, Just Signal

These aren't required, but strong candidates often do one or two:

- Retry with exponential backoff on LLM failures
- Confidence score on document classification
- n8n or Make workflow instead of (or alongside) raw code
- Docker or one-command local setup
- Async / batch processing for multiple documents

Submission Format

GitHub repo	Public repository with clean commit history — we should be able to clone and run it
README.md	Setup instructions, how to run, design decisions, known limitations — keep it under 1 page
sample_output/	Folder showing pipeline output (JSON extractions, routing log) on all 10 provided test files
Video walkthrough	3–5 min Loom or screen recording — REQUIRED. See details below.

Video walkthrough — what to cover (3–5 min):

1. Run the pipeline live on the 10 sample documents — we want to see it actually execute, not slides.
2. Show one successful Slack routing and one Sheet/CSV entry being written.
3. Trigger a deliberate error (e.g. feed it a blank file) and show how your agent handles it.
4. Briefly walk through one design decision you made — prompt structure, output validation, or how you handled the ambiguous invoice (inv_005).

Use Loom (free at loom.com), OBS, or any screen recorder. Share a public link in your README. Submissions without a video will not be reviewed.

We care more about what you've built than where you studied. Personal projects, coursework, and freelance work all count. Good luck.